

Assignment II - PHYS 708

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Problem 1

(a) Obtain the spectrum of particles accelerated by diffusive shock acceleration by solving the following transport equation* in the rest frame of the shock, where

$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial x} \left(p \frac{\partial f}{\partial p} \right) + Q(x, p) = 0,$$

x denotes the position with respect to the shock front (positive downstream and negative upstream),

u is the velocity of the gas (different for the different regions),

$D(p)$ is the diffusion coefficient assumed to be the same on both sides of the shock front,

$f(x, p)$ is the phase-space distribution function for particles with momentum p ,

$Q(x, p)$ is the phase-space density of thermal seed particles injected at the shock.

(b) Using the solution obtained, show that the particle spectrum follows a power law

$$N(p) \propto p^{-\Gamma}$$

with an index $\Gamma = 2$ for strong shocks.

Solution:

(a) We work in the rest frame of the shock, located at $x = 0$.

- Upstream region: $x < 0$, plasma speed u_1 .
- Downstream region: $x > 0$, plasma speed u_2 with $u_1 > u_2 > 0$.

The steady-state transport equation* governing particle motion is

$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial x} \left(p \frac{\partial f}{\partial p} \right) + Q(x, p) = 0.$$

We proceed by solving this one-dimensional differential equation in different regions under specific boundary conditions.

1 Solutions Away from the Shock

In each region (upstream or downstream), u is constant since it is either u_1 or u_2 . Thus $\partial u / \partial x = 0$. Also, far from the shock there are no sources, so $Q = 0$. The equation simplifies to

$$\frac{\partial}{\partial x} \left(D \frac{\partial f}{\partial x} \right) - u \frac{\partial f}{\partial x} = 0.$$

Integrating once with respect to x gives

$$D \frac{df}{dx} - uf = C(p),$$

where $C(p)$ is an integration constant that depends on momentum.

This is a first-order linear ODE. We rearrange and separate variables:

$$\frac{df}{dx} - \frac{u}{D} f = \frac{C(p)}{D}.$$

Multiplying through by the integrating factor $e^{-ux/D}$:

$$\frac{d}{dx} \left(f e^{-ux/D} \right) = \frac{C(p)}{D} e^{-ux/D}.$$

Integrating over x yields

$$f e^{-ux/D} = -\frac{C(p)}{u} e^{-ux/D} + A(p),$$

where $A(p)$ is an integration constant. Hence,

$$f(x, p) = A(p) e^{ux/D} - \frac{C(p)}{u}.$$

The general solution then becomes:

$$f(x, p) = A(p) + B(p) e^{ux/D(p)}.$$

Boundary conditions are applied as follows:

$$\begin{aligned} \text{Upstream } (x \rightarrow -\infty) : & \quad f_1(x, p) = f_s(p) e^{u_1 x / D(p)}, \\ \text{Downstream } (x \rightarrow +\infty) : & \quad f_2(x, p) = f_s(p). \end{aligned}$$

Upstream condition: $f = 0$ as there are no particles far upstream. Therefore, we take $A(p) = 0$ and leave the exponential term.

Downstream condition: To ensure that f is finite, we must take $B(p) = 0$.

Thus,

$$f(x, p) = \begin{cases} f_s(p) e^{u_1 x / D(p)}, & x < 0, \\ f_s(p), & x > 0. \end{cases}$$

We clearly see that $f_s(p) = f(0, p)$.

2 Shock Boundary Condition

We now solve the transport equation at the discontinuity $x = 0$, treating the shock as infinitesimally thin.

Assume

$$\frac{du}{dx} = (u_2 - u_1)\delta(x), \quad Q(x, p) = q_0(p)\delta(x), \quad q_0(p) = \frac{u_1 n_i}{4\pi p_i^2} \delta(p - p_i).$$

Define the flux in the x -direction:

$$J = -D \frac{df}{dx} + uf.$$

Then the equation can be written as

$$\frac{dJ}{dx} + \frac{1}{3} \frac{du}{dx} \left(p \frac{df}{dp} \right) + Q = 0.$$

Integrating across the shock from $-\epsilon$ to $+\epsilon$ and taking the limit $\epsilon \rightarrow 0$ gives

$$\lim_{\epsilon \rightarrow 0} \left\{ \int_{-\epsilon}^{+\epsilon} \frac{dJ}{dx} dx + \frac{1}{3} \int_{-\epsilon}^{+\epsilon} \frac{du}{dx} p \frac{df}{dp} dx + \int_{-\epsilon}^{+\epsilon} Q dx \right\} = 0.$$

$$\lim_{\epsilon \rightarrow 0} \left\{ \int_{-\epsilon}^{+\epsilon} \frac{dJ}{dx} dx + \frac{1}{3} \int_{-\epsilon}^{+\epsilon} (u_2 - u_1) \delta(x) p \frac{df}{dp} dx + \int_{-\epsilon}^{+\epsilon} q_0(p) \delta(x) dx \right\} = 0.$$

$$[J]_{-}^{+} + \frac{1}{3} (u_2 - u_1) p \frac{df}{dp} \Big|_{x=0} + q_0(p) = 0.$$

Substituting, we obtain the jump condition:

$$[J]_{-}^{+} + \frac{u_2 - u_1}{3} p \frac{df_s}{dp} + q_0(p) = 0.$$

We now compute each flux:

$$J_{-} = -D \left(\frac{u_1}{D} f_s \right) + u_1 f_s = 0,$$

$$J_{+} = -D(0) + u_2 f_s = u_2 f_s.$$

Hence,

$$[J]_{-}^{+} = J_{+} - J_{-} = u_2 f_s.$$

The final boundary condition becomes

$$u_2 f_s + \frac{u_2 - u_1}{3} p \frac{df_s}{dp} + q_0(p) = 0.$$

3 Solving for f_s by considering Momenta above Injection $p > p_i$

For $p > p_i$, $q_0(p) = 0$, thus

$$p \frac{df_s}{dp} = -\frac{3u_2}{u_2 - u_1} f_s.$$

Rewriting,

$$\frac{1}{f_s} \frac{df_s}{dp} = -\frac{3u_2}{u_2 - u_1} \frac{1}{p}.$$

Integrate with respect to p :

$$\ln f_s = -\frac{3u_2}{u_2 - u_1} \ln p + C, \quad f_s = Ap^{-\gamma},$$

where $A = e^C$ and $\gamma = 3u_1/(u_1 - u_2) = 3r/(r - 1)$.

4 Normalization

We finally consider integrating over a small region around p_i to find the normalization constant.

$$\lim_{\epsilon \rightarrow 0} \left\{ \int_{p_i - \epsilon}^{p_i + \epsilon} u_2 f_s dp + \int_{p_i - \epsilon}^{p_i + \epsilon} \frac{u_2 - u_1}{3} p \frac{df_s}{dp} dp + \int_{p_i - \epsilon}^{p_i + \epsilon} q_0(p) dp \right\} = 0$$

Evaluating each term:

1. f_s is continuous at p_i and therefore $\int_{p_i - \epsilon}^{p_i + \epsilon} u_2 f_s dp = 0$
2. $\int_{p_i - \epsilon}^{p_i + \epsilon} p \frac{df_s}{dp} dp \sim p_i (f_s(p_i^+) - f_s(p_i^-)) = p_i \cdot (Ap_i^{-\gamma} - 0) = Ap_i^{1-\gamma}$
3. $\int_{p_i - \epsilon}^{p_i + \epsilon} q_0(p) dp = \int_{p_i - \epsilon}^{p_i + \epsilon} \frac{u_1 n_i}{4\pi p} \delta(p - p_i) dp = \frac{u_1 n_i}{4\pi p_i^2}$

Thus,

$$\frac{u_2 - u_1}{3} Ap_i^{1-\gamma} = -\frac{u_1 n_i}{4\pi p_i^2},$$

which gives

$$A = \frac{3u_1}{u_1 - u_2} \frac{n_i}{4\pi} p_i^{\gamma-3} = \frac{\gamma n_i}{4\pi} p_i^{\gamma-3}.$$

The normalized shock distribution is

$$f_s(p) = f(0, p) = \frac{\gamma n_i}{4\pi p_i^3} \left(\frac{p}{p_i} \right)^{-\gamma}, \quad \gamma = \frac{3u_1}{u_1 - u_2} = \frac{3r}{r - 1}.$$

5 Energy Spectrum

(b) The omnidirectional energy (momentum) spectrum is

$$N(p) = 4\pi p^2 f(p) \propto p^{2-\gamma} = p^{-\Gamma}, \quad \Gamma = \gamma - 2 = \frac{r+2}{r-1}.$$

For a strong shock ($r = 4$):

$$N(p) \propto p^{-2}.$$